Nuclear Physics

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Prerequisite Knowledge

- Basic idea of electric charge and Coulomb's law
- Separable differential equations (see the Calculus handout — the decay law is derived using exactly this method)
- The random-walk / mean-free-path idea from the Thermodynamics handout is used, and rebuilt briefly, in the neutrino section

This handout covers four topics: *basic concepts including the structure of the atom; mass defect and binding energy; radioactivity; and neutrinos.* (The last of these is marked (Q) on the syllabus, meaning it's examined qualitatively rather than through heavy calculation — we'll still build real understanding, just without demanding derivations at the same depth as the rest.) These four topics tell one continuous story, and it happens to be the story of where every element in your body, and every watt of light a star produces, actually comes from. We'll build it from the atom upward: what a nucleus is, why some arrangements of protons and neutrons are bound more tightly than others, why that tight binding has a peak that fusion climbs toward and fission climbs toward from the other side, why unstable nuclei fall apart on a strict statistical schedule, and finally, why an almost-massless, almost non-interacting particle turned out to be one of the most important messengers we have for what's happening inside a star *right now*, rather than thousands of years ago.

1 Basic Concepts: The Structure of the Atom

Discovering That the Atom is Mostly Empty Space

At the start of the 20th century, the prevailing picture of the atom (J.J. Thomson's "plum pudding" model) imagined positive charge spread out smoothly through the whole atom, with electrons studded through it. Ernest Rutherford's team tested this by firing alpha particles (helium nuclei, positively charged) at a thin sheet of gold foil, expecting them to pass through with only slight deflections, since a smoothly spread-out charge shouldn't be able to exert a very concentrated force on anything.

What they found instead was startling: the overwhelming majority of alpha particles sailed straight through the foil essentially undeflected, but a small fraction — roughly 1 in 8,000 — bounced almost straight back, as if they'd hit something small, dense, and immovable. Rutherford himself described his own astonishment at this result, comparing it to firing a heavy shell at a piece of tissue paper and having it rebound directly back at the gun. The only way to explain both observations together — easy passage for most particles, violent rebound for a rare few — was if almost all of an atom's mass and positive charge is concentrated in a tiny central **nucleus**, with the rest of the atom being mostly empty space through which electrons orbit.

A Sense of Scale

If an atom were scaled up to the size of a large sports stadium, its nucleus would be about the size of a fly sitting at the center of the pitch — and yet that fly would contain essentially all of the atom's mass. This is worth holding onto as a mental picture: matter, at the scale we experience it, is overwhelmingly empty space, held together by electromagnetic forces acting across that emptiness.

Nuclear Notation and Nuclear Size

A nucleus is made of Z protons and N neutrons (collectively, **nucleons**), with mass number $A = Z + N$. We write a given nuclear species as ${}^{A}_{Z}X$, where X is the chemical symbol (itself determined entirely by Z, since Z is what defines which element it is). Nuclei sharing the same Z but different N are **isotopes** of the same element — chemically almost identical (since chemistry is governed by the electron cloud, which only cares about Z), but often very different nuclear properties.

Experimentally, nuclear radii follow a strikingly simple rule:

$$\boxed{R = r_0 A^{1/3}}, \qquad r_0 \approx 1.2\text{ fm } (1\text{ fm} = 10^{-15}\text{ m}).$$

This says nuclear *volume* ($\propto R^3 \propto A$) is directly proportional to the number of nucleons — in other words, **nuclear density is essentially constant**, the same for a light nucleus and a heavy one. Picture stacking oranges into a crate: each orange takes up the same amount of space regardless of how many oranges are already in the crate, so the crate's total volume just scales with the number of oranges. Nucleons behave the same way — each nucleon "claims" roughly the same fixed volume, and packing more of them in simply makes a proportionally bigger (but equally dense) nucleus. This single fact will be the seed of a real, derivable formula in the next section.

The Strong Nuclear Force

Protons, all carrying the same positive charge, should repel each other ferociously at the sub-femtometre distances inside a nucleus — so what holds a nucleus together at all? The answer is the **strong nuclear force**: an attractive force between nucleons (proton-proton, neutron-neutron, and proton-neutron alike) that is far stronger than the electromagnetic repulsion between protons, but only over an extremely short range, roughly the size of a nucleon itself (~ 1 fm). Beyond that tiny range, it essentially vanishes.

This short range has a profound consequence, worth sitting with now because it drives the entire next section: a given nucleon only feels the strong attraction from its immediate neighbors, no matter how large the nucleus is (this is called **saturation**). But the electromagnetic repulsion between protons has no such short range — every proton in the nucleus repels *every other* proton, no matter how far apart they are within the nucleus. As nuclei get bigger, the short-range attractive force stays "local," while the long-range repulsive force keeps accumulating across the whole nucleus. This tug-of-war between a short-range attraction and a long-range repulsion is exactly what determines how tightly bound a nucleus is — and it's the subject of the next section.

2 Mass Defect and Binding Energy

Mass Defect: Where Does the Missing Mass Go?

Here's a fact that seems to violate conservation of mass, until you remember Einstein: if you carefully weigh a nucleus, its mass is always *less* than the sum of the masses of its individual protons and neutrons, weighed separately. This missing mass is called the **mass defect**:

$$\Delta m = \left[Zm_p + Nm_n\right] - M_{\text{nucleus}}.$$

The resolution is Einstein's mass-energy equivalence, $E = mc^2$: mass and energy are two faces of the same thing, and the "missing" mass has been converted into the energy that binds the nucleus together, released when the nucleus was assembled from its separate parts. This is called the **binding energy**:

$$\boxed{BE = \Delta m\, c^2.}$$

Equivalently: binding energy is the energy you would need to supply to tear a nucleus apart into its separate, free protons and neutrons. A larger binding energy means a more tightly bound, harder-to-break-apart nucleus.

Building the Binding Energy From Physical Reasoning

Rather than just plotting binding energy against mass number and describing its shape, let's actually *derive* why it has that shape, piece by piece, from the physics in Section 1.

The volume term. Because the strong force saturates (each nucleon only really "feels" its immediate neighbors, regardless of how big the nucleus is), each nucleon contributes roughly the same fixed amount of binding energy, independent of A. Total binding energy from this effect alone is therefore directly proportional to the number of nucleons:

$$BE_{\text{volume}} = a_V A.$$

The surface term. The volume term over-counts slightly: nucleons sitting right at the *surface* of the nucleus have fewer neighbors than nucleons buried in the interior (much like how a person at the edge of a tightly packed crowd has fewer people pressed against them than someone in the middle), so they contribute less binding than the volume term assumes. This is a correction proportional to the nucleus's surface area. Since $R = r_0 A^{1/3}$ (Section 1), surface area scales as $R^2 \propto A^{2/3}$, giving a *negative* correction:

$$BE_{\text{surface}} = -a_S A^{2/3}.$$

This surface penalty matters proportionally much more for small nuclei (where a larger fraction of all nucleons sit on the surface) than for large ones — exactly like how a small pebble has a much higher surface-area-to-volume ratio than a boulder.

The Coulomb term, derived rigorously. Now for the repulsive piece: the electrostatic self-energy of the nucleus's total proton charge, spread through the nuclear volume. Let's actually derive this properly, using the same integration technique from the Calculus handout's "Applications of Integrals" section — building the charged sphere up one thin shell at a time.

Imagine assembling a uniformly charged sphere of total charge $Q = Ze$ and radius R by bringing in thin spherical shells of charge from infinity, one at a time. When the sphere has been built up to radius r (with charge $q(r)$ already in place, using uniform charge density $\rho = \frac{3Q}{4\pi R^3}$, so $q(r) = Q(r/R)^3$), the energy cost of bringing in the next thin shell $dq = \rho \cdot 4\pi r^2\, dr$ from infinity is

$$dU = \frac{kq(r)}{r}\, dq = \frac{k}{r} \cdot Q \left(\frac{r}{R}\right)^3 \cdot \rho\, 4\pi r^2\, dr = \frac{4\pi k\rho Q}{R^3}\, r^4\, dr.$$

Integrating over the whole sphere, from $r = 0$ to $r = R$:

$$U = \frac{4\pi k\rho Q}{R^3} \int_0^R r^4\, dr = \frac{4\pi k\rho Q}{R^3} \cdot \frac{R^5}{5} = \frac{4\pi k\rho Q R^2}{5}.$$

Substituting $\rho = \frac{3Q}{4\pi R^3}$:

$$\boxed{U = \frac{3}{5} \cdot \frac{kQ^2}{R}.}$$

This is a clean, fully rigorous result: the electrostatic self-energy of a uniformly charged sphere. For a nucleus, one small refinement: a proton doesn't repel *itself*, so the correct count of repelling pairs among Z protons is $Z(Z-1)$, not Z^2 (there are $Z(Z-1)$ ways to pick an ordered pair of two different protons). Making this replacement, and substituting $R = r_0 A^{1/3}$:

$$BE_{\text{Coulomb}} = -\frac{3ke^2}{5r_0} \cdot \frac{Z(Z-1)}{A^{1/3}} \equiv -a_C \frac{Z(Z-1)}{A^{1/3}}.$$

Let's check this against reality by actually computing a_C from first principles:

$$a_C = \frac{3ke^2}{5r_0} = \frac{3(8.99 \times 10^9)(1.6 \times 10^{-19})^2}{5(1.2 \times 10^{-15})} \approx 1.15 \times 10^{-13}\ \text{J} \approx 0.72\ \text{MeV}.$$

This matches the experimentally fitted value of $a_C \approx 0.71$ MeV extremely well — a genuine, first-principles derivation of a real nuclear physics constant, straight out of freshman electrostatics and one clean integral.

Why the Volume and Surface Terms Aren't Derived the Same Way

Unlike the Coulomb term, the volume and surface coefficients ($a_V \approx 15.8$ MeV, $a_S \approx 18.3$ MeV) can't be derived from a clean classical formula the way electrostatic self-energy can — they depend on the detailed, genuinely quantum-mechanical behavior of the strong force, and are instead fitted to match real measured nuclear masses. This combination — some terms derived exactly, others fitted empirically — is exactly why the resulting formula (the **semi-empirical mass formula**, or Bethe-Weizsäcker formula) carries the word "semi-empirical" in its name.

The Binding Energy Curve, and Why It Has a Peak

Combining these pieces (there are two more small correction terms in the full formula — a pairing term and an asymmetry term — which we'll skip for this level of treatment), the binding energy per nucleon, BE/A, can be plotted as a function of A:

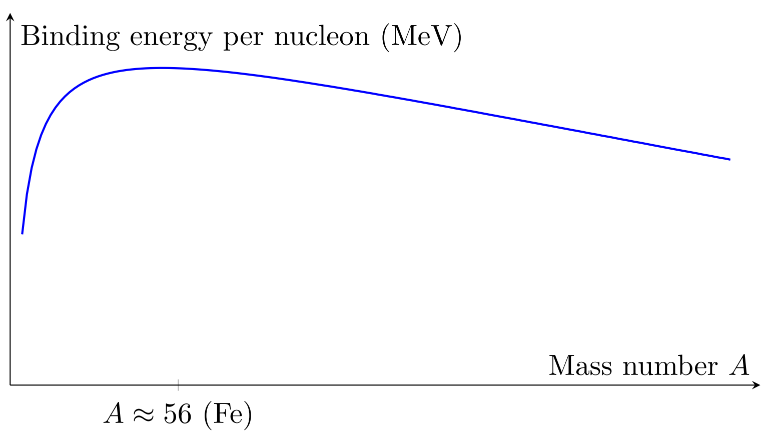

Figure 1: Binding energy per nucleon vs. mass number, from the derived formula above (using $Z \approx A/2$ for illustration). The peak sits right around iron ($A \approx 56$), matching the real curve closely — this smooth liquid-drop trend misses some sharp spikes for very light, tightly-bound nuclei like ^{4}He, ^{12}C, and ^{16}O (a shell-structure effect beyond this model), but the overall shape and peak location are correctly captured.

The shape makes complete physical sense given what we derived: for small A, the surface term (a proportionally large penalty for small nuclei) dominates the competition, so binding energy per nucleon *rises* as A increases and the surface-to-volume ratio improves. But for large A, the Coulomb term — growing roughly as $Z^2/A^{1/3} \sim A^{5/3}$ for $Z \propto A$ — eventually outpaces the volume term (which only grows as A), so binding energy per nucleon *falls* again for the heaviest nuclei. The peak, where these two competing trends balance, sits right around iron and nickel.

Stellar Application: Why Stars Fuse Light Elements, and Why Fusion Stops at Iron

This single curve is the entire reason stars shine, and the entire reason they eventually stop shining the way they started.

Fusion releases energy whenever it moves nuclei *up* this curve — combining light nuclei (sitting low on the curve, on its steep left flank) into a heavier nucleus that sits higher up, closer to the peak. The mass difference between "before" and "after" is released as energy, via $E = \Delta m\, c^2$ — this is exactly what powers the Sun, fusing hydrogen into helium and climbing a long way up that steep left-hand slope.

Fission releases energy whenever it moves nuclei *up* this curve from the *other* direction — splitting a very heavy nucleus (sitting on the shallow right-hand decline, past the peak) into two medium-mass fragments that individually sit higher on the curve than the original heavy nucleus did. This is the reverse process, but it works for the same underlying reason: moving toward the peak always releases energy, whether you approach it from the light side by fusing, or from the heavy side by splitting.

And this is exactly why fusion inside a star has a hard stop at iron. Once a star's core has fused its way up to iron and nickel — sitting right at the peak of the curve — there is *no further nuclear reaction, fusion or fission, that can extract net energy.* Any attempt to fuse iron into something heavier would require *adding* energy, not releasing it, since you'd be moving back down the curve's declining right-hand slope. When a massive star's core finally becomes predominantly iron, it has run out of nuclear fuel in the most fundamental sense possible — not because the raw material has run out, but because the entire energy landscape available to nuclear reactions has been fully climbed. This is precisely the trigger for core collapse and a Type II supernova, and it's why we'll come back to iron-peak elements again in the neutrino section.

Elements heavier than iron, which are very much real and make up a meaningful fraction of the periodic table (including gold, platinum, and uranium), simply cannot be produced by ordinary fusion at all — they require a completely different formation mechanism, which is exactly where Section 3's radioactivity comes back into the story.

3 Radioactivity

Why Nuclei Decay

Not every combination of protons and neutrons is stable. Plotting stable nuclei on a chart of N versus Z reveals a narrow "valley of stability": light stable nuclei cluster near $N = Z$ (equal numbers of protons and neutrons), but as Z grows, the accumulating Coulomb repulsion (Section 2) requires progressively *more* neutrons than protons to stay bound (neutrons add strong-force attraction without adding Coulomb repulsion). Nuclei that stray from this valley — too many protons, too many neutrons, or simply too heavy overall — are unstable, and spontaneously transform into more stable configurations via **radioactive decay**.

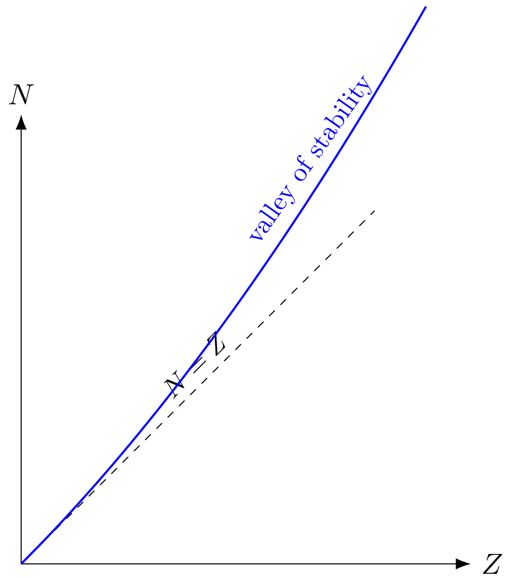

Figure 2: Schematic chart of nuclides. Light stable nuclei sit near $N = Z$; heavier stable nuclei need progressively more neutrons than protons, since neutrons add binding without adding Coulomb repulsion. Nuclei off this valley are unstable.

The three main decay modes each nudge an unstable nucleus back toward this valley in a different way:

Alpha decay: the nucleus ejects a ${}^4_2\text{He}$ nucleus (two protons, two neutrons), reducing both Z and N by 2. This is common for very heavy nuclei, where shedding a compact, tightly bound chunk (helium-4 sits on a local peak of the binding curve, from Section 2's shell-effect caveat) relieves Coulomb strain efficiently.

Beta decay: a neutron converts to a proton (β^- decay, $n \to p^+ + e^- + \bar{\nu}_e$), or a proton converts to a neutron (β^+ decay, $p^+ \to n + e^+ + \nu_e$), moving the nucleus horizontally on the chart above, toward whichever side of the valley it's missing. We'll come back to that second particle appearing in these equations — the neutrino — in Section 4.

Gamma decay: after an alpha or beta decay, the resulting nucleus is often left in an excited (higher-energy) configuration, and relaxes to its ground state by emitting a high-energy photon, with Z and N both unchanged — exactly analogous to an electron dropping between atomic energy levels (Electromagnetic Theory & Quantum Physics handout), just at nuclear energy scales, roughly a million times larger.

Deriving the Decay Law

Here's the key physical assumption behind radioactive decay, and it's a statement about probability, not about any individual nucleus: each unstable nucleus has a fixed, constant probability per unit time, λ (the **decay constant**), of decaying — completely independent of how old it is, or what its neighboring nuclei are doing. (This is genuinely different from, say, a light bulb, which becomes *more* likely to fail the longer it's already run; a radioactive nucleus has no such memory.)

If there are N such nuclei present at some instant, the expected number of decays in a small time interval dt is simply $\lambda N\, dt$ (each of the N nuclei independently contributing a $\lambda\, dt$ chance of decaying), so

$$dN = -\lambda N\, dt \quad \Longrightarrow \quad \frac{dN}{dt} = -\lambda N.$$

This is a separable differential equation, solved by exactly the method from the Calculus handout:

$$\int \frac{dN}{N} = -\lambda \int dt \quad \Longrightarrow \quad \ln N = -\lambda t + C \quad \Longrightarrow \quad \boxed{N(t) = N_0 e^{-\lambda t}.}$$

The number of undecayed nuclei falls off exponentially, with no finite time at which it reaches exactly zero — there's always some (ever-shrinking) chance a few nuclei survive arbitrarily long.

Two useful timescales follow immediately. The **half-life** $T_{1/2}$ is the time for half the sample to decay, $N(T_{1/2}) = N_0/2$:

$$\frac{1}{2} = e^{-\lambda T_{1/2}} \quad \Longrightarrow \quad \boxed{T_{1/2} = \frac{\ln 2}{\lambda}.}$$

The **mean lifetime** $\tau = 1/\lambda$ is the average time an individual nucleus survives before decaying (a short calculus exercise using $\tau = \frac{1}{N_0}\int_0^\infty t\,|dN|$ confirms this), giving the simple relation $T_{1/2} = \tau \ln 2$.

Worked Example — Radium's Half-Life (NSEA style)

One gram of Radium (atomic weight 226) emits 4×10^{10} particles per second. Find its half-life. The number of atoms in 1 gram is $N = \frac{1}{226} \times (6.022 \times 10^{23}) \approx 2.665 \times 10^{21}$. Since activity $= \lambda N$:

$$\lambda = \frac{4 \times 10^{10}}{2.665 \times 10^{21}} \approx 1.50 \times 10^{-11}\ \text{s}^{-1} \quad \Longrightarrow \quad T_{1/2} = \frac{\ln 2}{\lambda} \approx 4.62 \times 10^{10}\ \text{s}.$$

Worked Example — Mean Life From a Fractional Loss (NSEA style)

A radioactive sample loses $\frac{7}{8}$ of its initial amount in 4.606 days. Find the mean life. Losing 7/8 means $N/N_0 = 1/8 = 2^{-3}$, so exactly 3 half-lives have passed: $4.606 = 3T_{1/2} \Rightarrow T_{1/2} \approx 1.535$ days. Then

$$\tau = \frac{T_{1/2}}{\ln 2} \approx \frac{1.535}{0.693} \approx 2.2\ \text{days}.$$

Stellar Application: Building the Elements Heavier Than Iron

We left off Section 2 with a puzzle: fusion can't build anything past the iron peak, so where do gold, platinum, and uranium actually come from? The answer is radioactivity, working in partnership with a completely different nuclear process: **neutron capture**. A nucleus can absorb a free neutron (no Coulomb repulsion to overcome, since neutrons are uncharged) to become a heavier isotope of the same element, and if that new isotope is unstable, it beta-decays — a neutron converting to a proton — bumping the nucleus to the *next* element up the periodic table, all without ever requiring the enormous energy cost that fusing past the iron peak would demand.

This happens in nature in two very different regimes. The **s-process** ("slow") occurs inside evolved stars (particularly AGB stars) where neutron densities are modest enough that a freshly captured neutron usually has time to beta-decay, if it's going to, before the next capture happens — building up heavier elements slowly, over thousands of years, one careful step at a time. The **r-process** ("rapid") occurs in far more extreme environments — core-collapse supernovae and neutron star mergers — where neutron densities are so extreme that a nucleus can capture many neutrons in rapid succession, far faster than beta decay can keep pace, before finally beta-decaying back toward stability once the neutron flux subsides. The r-process is responsible for roughly half of all elements heavier than iron, including essentially all of the universe's gold and platinum — meaning that mass defect, binding energy, and radioactive beta decay, together, are the actual origin story of some of the most prized elements on Earth.

4 Neutrinos (Q)

The Particle Invented to Save a Conservation Law

Here's a genuinely dramatic story in the history of physics. In beta decay, a nucleus emits an electron, and if that were the whole story — nucleus decays into a slightly lighter nucleus plus one electron, a clean two-body process — then energy and momentum conservation would force the emitted electron to always carry away one single, precisely fixed energy for any given decay (just as it does in alpha decay). But when physicists in the 1920s actually measured the energies of electrons emitted in beta decay, they found something bewildering: a broad, continuous range of energies, not a single fixed value.

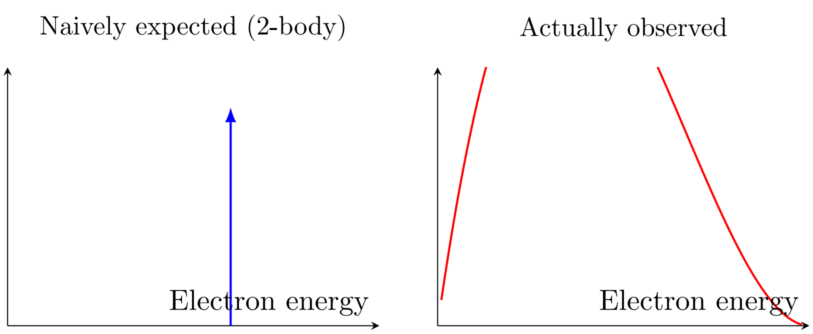

Figure 3: The beta-decay puzzle. A clean two-body decay predicts one fixed electron energy (left); real measurements showed a continuous spread (right) — as if energy itself were sometimes going missing.

This looked, alarmingly, like energy conservation might simply be violated in beta decay — a few prominent physicists at the time were seriously willing to entertain that possibility. Wolfgang Pauli instead proposed, in 1930, what he himself called a "desperate remedy": that a third, invisible particle was also being emitted in every beta decay, carrying away a variable share of the energy and momentum, in a way that always keeps the books balanced but is different every single time — which is exactly why the electron's own share looks continuous rather than fixed. This hypothetical particle needed to be electrically neutral (for charge to balance) and needed to interact so weakly with matter that it had evaded direct detection entirely. Enrico Fermi later named it the **neutrino** ("little neutral one"), and it wasn't directly detected until 1956 — a full 26 years after Pauli first proposed it purely to rescue a conservation law on paper.

Why Neutrinos Are So Hard to Catch

Neutrinos interact only via the weak nuclear force (and gravity, negligibly), which is exactly what makes them so ghostly. Their mean free path through ordinary matter is staggering — a typical solar neutrino could pass through several *light-years* of solid lead before being likely to interact even once.

This is worth contrasting directly with the photon random walk you derived in the Thermodynamics handout. A photon born in the Sun's core has a mean free path of order centimetres, and takes somewhere from thousands to tens of thousands of years to randomly walk its way out. A neutrino born in that same core essentially never interacts again on its way out — it simply travels in a straight line, at (very nearly) the speed of light, and reaches the Sun's surface, and then Earth, in about the same eight minutes light takes to cover that distance in vacuum, once outside the Sun.

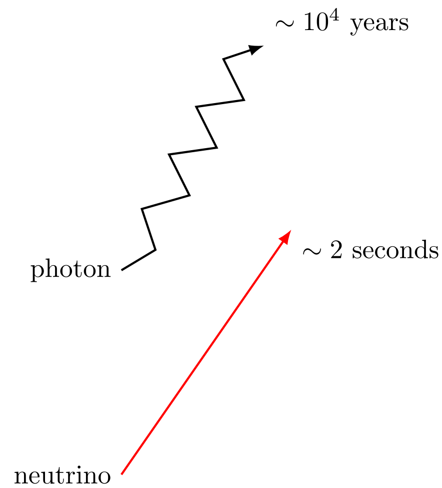

Figure 4: The same journey out of the Sun's core, for a photon (random walk, thousands of years) versus a neutrino (essentially a straight line, seconds).

This means solar neutrinos are an almost unbelievably direct messenger: **the light reaching your eyes today tells you what the Sun's core was doing thousands of years ago, but the neutrinos passing through your body right now (harmlessly, by the trillions, every second) tell you what the Sun's core is doing within the last few seconds.** Detecting and counting solar neutrinos is, quite literally, a real-time readout of the nuclear fusion rate happening in the Sun's core at this exact moment.

Stellar Application: The Solar Neutrino Problem

When physicists first built detectors sensitive enough to actually count solar neutrinos, starting in the 1960s, they found something puzzling: only about a third of the neutrino flux predicted by solar fusion models was actually being detected. This "solar neutrino problem" persisted for decades, and cast genuine doubt on either our understanding of the Sun's core, or our understanding of the neutrino itself.

The resolution, confirmed decisively by the Sudbury Neutrino Observatory in 2001, was the latter: neutrinos come in three types ("flavors"), and a neutrino created as one flavor can spontaneously change into another flavor while in flight, a genuinely quantum-mechanical effect called **neutrino oscillation** (which also tells us, incidentally, that neutrinos cannot be perfectly massless, since the oscillation mechanism requires it). Earlier experiments had only been sensitive to one particular flavor, so they were only ever catching about a third of the total — exactly matching the shortfall. The Sun's fusion models had been right all along; it was our picture of the neutrino that needed updating.

Stellar Application: A Supernova's Neutrino Burst Arrives Hours Before Its Light

Here is perhaps the single most spectacular confirmation of everything in this handout, observed directly: Supernova 1987A. When a massive star's iron core finally collapses (Section 2's inevitable endpoint once fusion reaches the iron peak), the collapse itself releases an enormous burst of neutrinos, created as protons and electrons are crushed together into neutrons throughout the collapsing core — essentially the same $p^+ + e^- \to n + \nu_e$ reaction from Section 3's beta decay discussion, happening all at once, everywhere in the core, in a fraction of a second.

Because neutrinos barely interact with matter at all, this entire burst streams straight out of the collapsing star almost instantly. But the visible light from the resulting explosion has to fight its way out through the star's own dense outer layers first — not quite the millennia-long journey of an undisturbed stellar core (Thermodynamics handout), but still a real delay, of order hours, before the shockwave reaches the surface and the supernova becomes optically visible.

When Supernova 1987A occurred in the Large Magellanic Cloud, neutrino detectors on Earth (several, independently) registered a burst of neutrinos roughly **three hours before** astronomers observed the corresponding brightening in visible light — a direct, real-world confirmation both that core collapse produces a neutrino burst, and that neutrinos genuinely do escape a collapsing star far faster than light does.

Worked Example — Counting Neutrinos From a Neutron Star's Birth (NSEA style)

When a $1.4\,M_\odot$ iron core collapses into a neutron star, roughly every proton present gets converted to a neutron via $p^+ + e^- \to n + \nu_e$, each conversion releasing one neutrino. Using iron's proton-to-nucleon ratio ($Z/A = 26/56$) as a stand-in for the core's initial composition, and $m_p \approx 1.67 \times 10^{-27}$ kg:

$$N_{\text{nucleons}} = \frac{1.4 \times (2 \times 10^{30})}{1.67 \times 10^{-27}} \approx 1.68 \times 10^{57}, \qquad N_\nu \approx \frac{26}{56} \times N_{\text{nucleons}} \approx 7.8 \times 10^{56}.$$

This simple bit of bookkeeping — no detailed weak-interaction theory required, just tracking how many protons need converting — lands right at the same order of magnitude actually released in a real core-collapse event, and shows why neutrino counting is such a powerful, if qualitative (hence the (Q) on this topic), tool for understanding what happens in the last second of a massive star's life.

5 Summary Sheet

Topic	Key Result
Nuclear radius	$R = r_0 A^{1/3}$, $r_0 \approx 1.2$ fm (constant nuclear density)
Mass defect / binding energy	$BE = \Delta m\, c^2$
Coulomb self-energy (derived)	$U = \frac{3}{5}\frac{kQ^2}{R}$; nuclear term $a_C \approx 0.72$ MeV
SEMF (schematic)	$BE = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}}$
Peak of BE/A	near $A \approx 56$ (iron/nickel) — fusion and fission both climb toward it
Decay law (derived)	$\frac{dN}{dt} = -\lambda N \Rightarrow N(t) = N_0 e^{-\lambda t}$
Half-life / mean life	$T_{1/2} = \ln 2/\lambda$; $\tau = 1/\lambda = T_{1/2}/\ln 2$
Heavy-element formation	s-process (slow) and r-process (rapid) neutron capture + β^- decay
Neutrino origin	Proposed by Pauli (1930) to save energy/momentum conservation in β decay
Neutrino escape time (Sun)	$\sim$ seconds, vs. $\sim 10^4$ years for photons (contrast: mean free path)
SN 1987A	Neutrino burst arrived ~ 3 hours before visible light